

L8

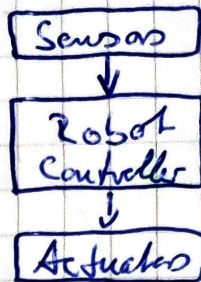
Behavior-based Robotics

• Classic Robot: Move $\rightarrow$ Think $\rightarrow$ Next Move

• Functional Decomposition

$\rightarrow$ Classical Approach: Top-Down Decomposition into logical units

- | | |
|----------------------------|--------------|
| 1) Processing sensory data | (Perception) |
| 2) Creation of world model | (Modelling) |
| 3) Planning | (Planning) |
| 4) Plan execution | (Execution) |



$\rightarrow$ Advantages:

- (+) System is precise
- (+) controllable
- (+) predictable

$\rightarrow$ Disadvantages:

- (-) difficult to handle noise and uncertainty
- (-) unit failure $\Rightarrow$ system failure
- (-) Computational expensive
- (-) granularity problem (model)

Braitenberg Vehicles

- Simple Agents:
 - Sensors: measure stimulus at given point
 - Actuators: transform input values into motion

eg: light sensor, more light = faster

Fear (a): Drive fast away from light then slows down (with distance)

Aggression (b): Drive fast to light, hitting light at full speed

Connection can also be inhibitory, i.e. stronger stimulus → smaller input to actuators

eg: Behavior, both "like" the source

Admires (a): Gets slower as it approaches the source until it stops

Explores (b): Slows down when near a source, but turns away from it looking out for stronger source

Different Stimuli (eg: temperature, oxygen)

→ vehicles will exhibit complex and dynamic behavior

→ Behavior:

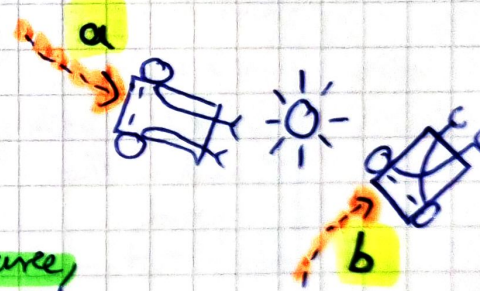
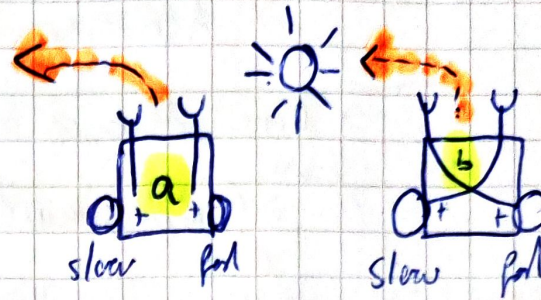
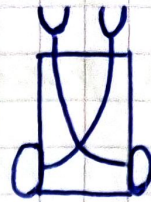
→ is goal-directed, fast, flexible and adaptive

→ might even appear intelligent

→ No cognitive processes

→ Agent is purely reactive

→ Simple architecture leads to robustness



Behavior-based Architecture

→ Brooks assumptions:

- Complex behavior does not require complex control system
- Simple to increase stability and robustness
- Robots should be autonomous to survive long without human
- Environment is 3D
- Absolute coordinate system leads to cumulative errors

→ Core ideas of a behavior-based system:

→ Embodiment: Robot has a real body: The body and the internal state of this body influence higher cognitive functions and the resulting behaviors.

physical interaction through body

→ Situatedness (Embeddness):

Robot operates directly with and in the world.

Robot is in a real situation: Robot must deal with sensory and motor eventualities where it operates and not with abstract descriptions of the world, i.e. physical interaction between the body and the world strongly constrain the possible behavior.

→ Stimuli relevant to behaviors $S = [s_1, \dots, s_n]$

→ Set of behaviors $B = [b_1, \dots, b_n]$

→ Weights/gains primitive behaviors $G = [g_1, \dots, g_n]$

→ Response of behavior i : $r_i = b_i(s_i)$

→ Overall response: $p = C(G * B(S))$

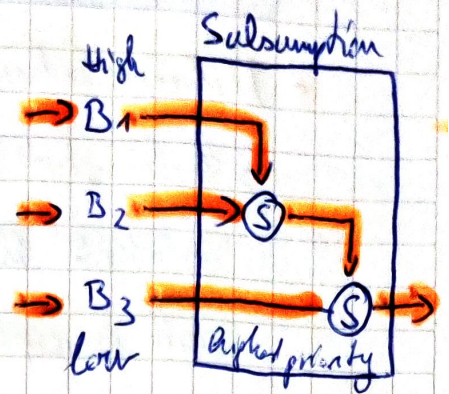
// vector of r_i eg $r_i = (\text{magnitude}, \text{angle})$

↳ Select / combine outputs to produce single response vector

Competitive Behavior Selection

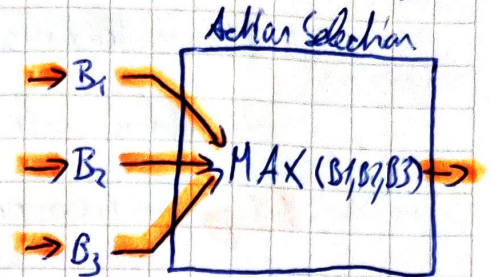
→ Subsumption:

Higher level behaviors can override low level behavior output



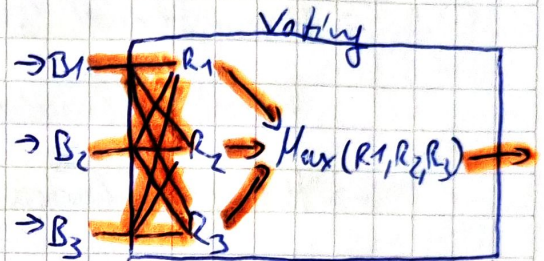
→ Action selection:

Selection through contention (eg signal strength)



→ Voting:

Behaviors vote for action response R



Subsumption Architecture (Brooks)

→ Decompose robot control in terms of task-achieving behaviors rather than functional modules (functional decomposition)

Identify
↓
Happiness
↓
Explore
↓
Wander
↓
Avoidance

→ Complex behavior may simply be the reflection of a complex environment

→ Layered

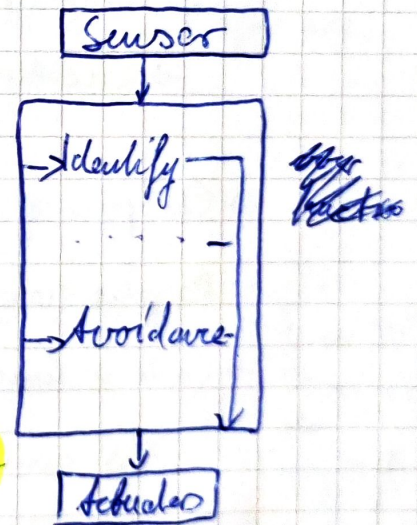
→ Higher level implies more specific behavior

→ All layers have access to sensors and outputs of lower level modules

→ Higher level can inhibit or resume output of lower levels or reset state of module

→ Bottom-Up Development (Next level only when last level works)

→ Widely used in real-time AI



28

Motor Schemas

→ Behavior specification for navigation of mobile robots: Multiple concurrent processes associated with perceptual schemas

→ Motor schemas output vector containing direction and strength (vector)

→ Eg for cooperative behavior selection (→ behavior based)

→ Perceptual Schema: (preprocessing for MS)

Provide environmental info specific for motor schema

→ Motor Schema:

→ Sensory input, navigation vector as ^{output} input

→ Can have internal state

→ Output of schemas can be visualised as vector/potential field

→ Response vector contains direction and strength

→ Each schema can be defined independently

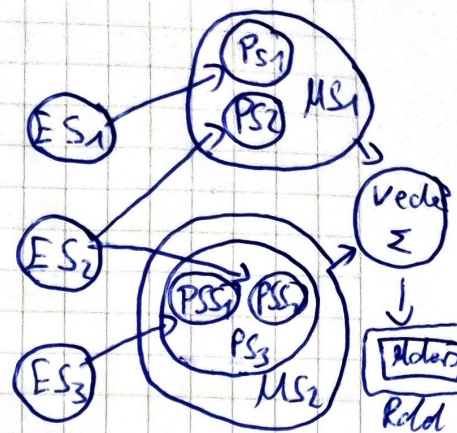
→ Schemas are combined in linear combination to single output vector

→ Two kinds of fields: attractive & repulsive

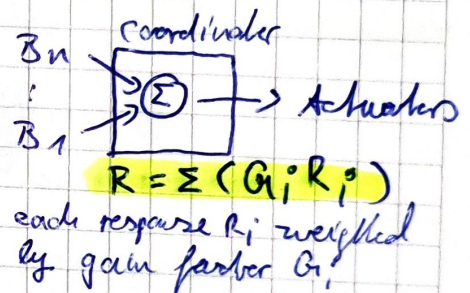
At each point:

$$U_{total} = U_{attraction} + U_{repulsion}$$

⇒ movement predetermined



PS: Perceptual Schema
PSS: Perceptual Subschema
MS: Motor Schema



• Schema example

→ Towards - Goal:

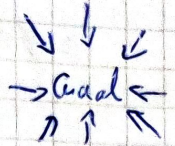
→ Ballistic: $V_{\text{magnitude}} = \text{fixed gain value}$

$V_{\text{direction}} = \text{towards goal}$

→ Guarded:

$$V_{\text{magnitude}} = \begin{cases} G, & \text{for } d \geq S \\ G \cdot \frac{d}{S}, & \text{for } d < S \end{cases}$$

$V_{\text{direction}} = \text{towards goal}$



$d = \text{distance to goal}$

$S = \text{radius of influence}$

→ Avoid - Obstacle:

$$V_{\text{magnitude}} = \begin{cases} 0, & d > S \\ \frac{S-d}{S-R} \cdot G, & R < d \leq S \\ \infty, & d \leq R \end{cases}$$

$G = \text{Gain}$

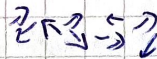
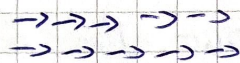
$d = \text{distance to object}$

$R = \text{Radius of Object}$

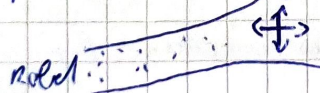
$S = \text{Sphere of Influence}$



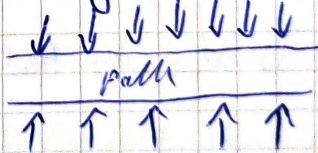
→ Simple: Move - Ahead, Noise



→ More complex: Dodge



Stay - On - Path

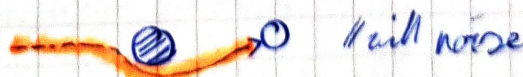


don't let
Towards it
the
stay on path

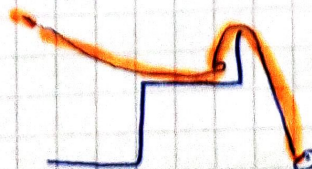
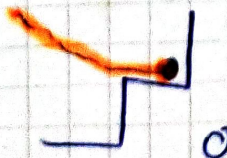
→ Overall robot behavior is integration of all active schemas (e.g., Linear Combination)

• Problems with schemas

Local Minima → Possible solution: Noise Schema



Cyclic Behavior → Possible solution: Avoid - Path Schema



1 on recently visited

Human-like Social Robots

→ Neuro-Inspired Companion (NICO)

→ Multimodal Research Platform:

Anthropomorphism, Scaled to realistic Env, Human-like response, Bimanual manipulation, Locomotion, Modular open source design, Adaptable to individual experimental setups, Affordable

→ NICO's Hands:

→ Artificial fingertips with deformation and vibration sensing

→ Haptic sensors for grasping and recognizing objects

→ NICO's Head:

→ child like appearance ⇒ avoids uncanny-valley effect

→ Sensors (2 cameras, 2 microphones) 

→ Emotion Display  

→ Bio-Inspired Development of Visuo-Motor Abilities

Developmental learning from environment interaction

Goal: Unified neural architecture for object picking

Fusing vision, motor and semantics

→ Object Picking Task:

Needed Components: Identify Object,

Locate Object in Space,

Compute and execute motor commands

Object Picking: Spectrum of approaches: Modular vs end-to-end

Modular approaches:

- Obj detection
- Pose estimation
- Inverse kinematics

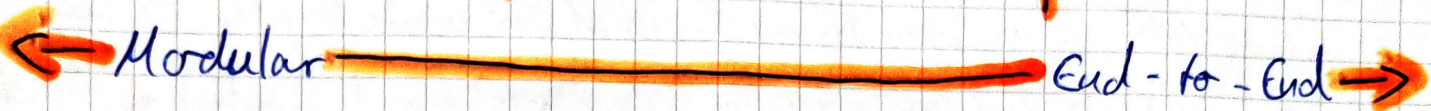
Neural end-to-end grasping:
single object grasping

Two-stage approach:

- Neural object detection
- Non-neural attention focus
- Neural end-to-end grasping

One-stage approach:

- Neural object picking
- Unified Arch
- End-to-End learning of object loc, direction and required kinematics



Bior-inspired Object-Picking Arch

Architecture Components:

- Visual processing: → Low-level visual processing
→ Pattern & Shape Processing

Low-level Hierarchical Visual Processing



Pattern & Shape Processing



Task-relevant Spatial Information

Motor Control

Goal Intention Encoding



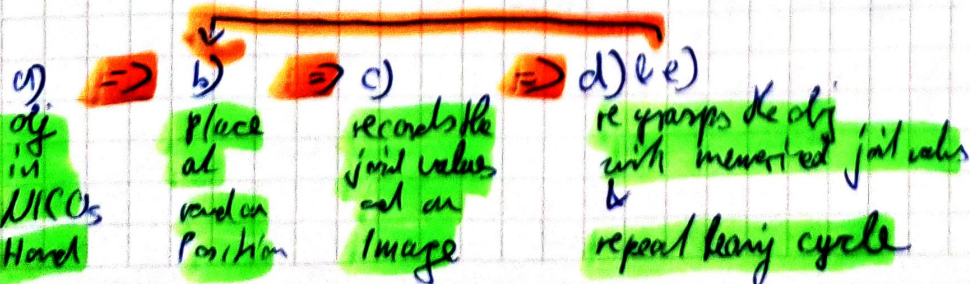
- Goal encoding
- Visuomotor processing: → Task-relevant spatial info
→ Motor control

Neural Grasp Learning: Self-Learning Cycle (Image, Joint (angle))

End-to-end learning of sensorimotor skills for obj grasping

1) Gather samples linking object positions to motor actions by placing an object at random positions

2) Train deep neural end-to-end network with inverted samples for grasping



• Data Augmentation

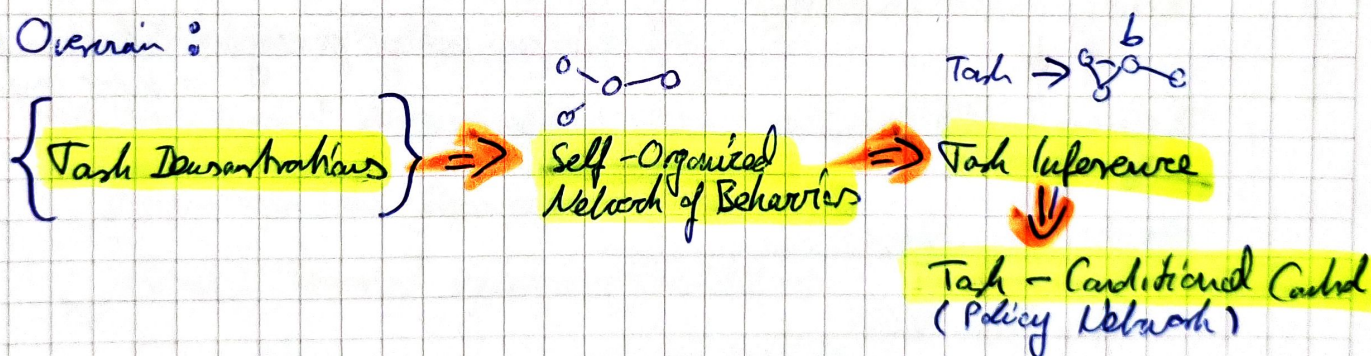
- Randomized subsets of distractor objects
- Semantic info of target object (eg: red, round, banana)

• Unsupervised Task Inference in Continual Robot Learning

→ Motivation:

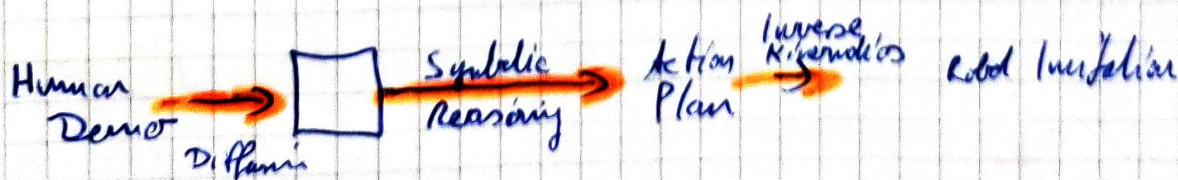
- Learning predefined tasks → learning a growing set of tasks
- To learn new tasks, the robot first needs to infer the task at hand
- Fixed task representations (eg labels) are incompatible with continual learning
- Existing approaches require extensive exploration to infer a task before executing the learned policy
- Observing a Demonstration

→ Overview:



• Imitation Learning

- Task demonstrations are shown to the robot and recorded
- From the examples, objects and transitions are estimated
- Robot infers inverse kinematics to perform tasks



Behavior-based agents

Advantages:

- (+) → Fast Reaction
 - Immediate mapping of sensory info onto motor actions
- (+) → Robustness
 - If one part fails, robot may retain some behavior ^{competence}
- (+) → Multiple Goals
 - Can follow multiple goals simultaneously by carrying ^{behavior}
- (+) → Extensibility
 - Easy to add new parts as top
- (+) → Simplicity
 - Complexity derives from continuous interaction of simple modules with env and each other
- (+) → Computational tractability
 - Usually simple calculations, parallelizable

Disadvantages:

- (-) → Limited Information
 - Agents without env models must have sufficient info from local env
- (-) → Non-local Information
 - How does agent take into account non-local info
- (-) → Learning Globally
 - Difficult to make reactive agent that learns globally
- (-) → Complex Dynamics
 - Hard to engineer agents with large number of behaviors